Reusable compute settings for GTSfM

Save a setup, reload it, and see what workers actually ran.

WHERE THIS FITS

Photos
Your input images

→

Studio setup
NEW: saved profiles + validation

→

Existing SfM
Dask schedules CPU/GPU workers

→

3D model
Cameras + sparse points

GTSfM turns photos into a 3D model. Dask assigns the work to CPU/GPU workers.

BEFORE: setup lived in the form

Custom worker, memory and GPU settings lived in the current form. No named setup to reload after restart.

AFTER: a reusable named profile

Save a profile. Reload it, validate it, launch through the existing runner, and compare requested vs. observed settings.

Makes experiments easier to repeat and configuration mistakes easier to spot.

CHECK: DID THE RECONSTRUCTION CHANGE?

Manual settings - 2 runs

Saved profile - 2 runs

→

Same RTX 3080
Same images and SfM configuration

→

EXACT FILE MATCH

Same exported files.

4 cameras | 1,509 tracks

Manual settings and saved profiles produced byte-identical reconstruction exports.

THE GAIN: repeatable setup + visible execution

Tested on four images with one RTX 3080. Speed, accuracy and larger-scene gains remain unproven.

Also checked: restart persistence, invalid-setting rejection and 64 focused tests.

Detailed methods and raw results: docs/benchmarks/raw/README.md

Evidence revision: 8dafa2de | Engineering repeatability result